

CV - **Katrine Alice Glasscock** (She/her)

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Research positions

- 08/2025 – current **Postdoctoral Fellow** at University of Oslo, Norway
Working in the Commander/Cosmoglobe group on Bayesian foreground modelling for CMB and CIB.
- 03/2025 – 08/2025 **Postdoctoral Researcher (PDRA)** at University of Manchester, UK
Working on MeerKAT/MeerKLASS calibration pipeline and data processing. EDI work, planning and hosting EDI session at National Astronomy Meeting.

Education and qualification

- 09/2021 – 05/2025 **PhD; Cosmology**, University of Manchester (First year at Queen Mary University of London)
Thesis title: *Bayesian estimation of diffuse radio foregrounds with large interferometric arrays*. Supervised by Phil Bull.
- 08/2018 – 10/2020 **MSc; Astronomy**, Aarhus University
ECTS GPA for Master's: 4.6. Thesis received highest mark.
Master's thesis: *21-cm Emission in the Early Universe in Non-Standard Cosmological Models*. Supervised by Steen Hannestad and Thomas Tram.
- 08/2015 – 06/2018 **BSc; Physics with Astronomy**, Aarhus University
ECTS GPA: 3.73. Thesis received second highest mark.
Bachelor's project: *Investigating the Dark Web – Cosmological N-body Simulations of Dark Matter & Dark Energy*. Supervised by Steen Hannestad.

Published/submitted works

My research focuses on the early Universe and large-scale structure through robust statistical modelling and observations. Research topics include: Bayesian modelling and component separation techniques for cosmological datasets. Foregrounds to the CMB and cosmological 21cm signal, with extensions to the cosmic infrared background (CIB) and zodiacal dust emission. Global 21cm experiments and excess radio backgrounds as probes of astrophysics and potential new physics.

11. Cosmoglobe DR2. VI. Disentangling hot and cold thermal dust emission with Planck HFI.
R. M. Sullivan, E. Gjerløw, M. Galloway, et al. (Submitted) A&A, 2026, [arXiv:2601.10640](#)
10. Cosmoglobe DR2. IV. Modelling starlight in DIRBE with Gaia and WISE.
M. Galloway, E. Gjerløw, M. San, et al. (Submitted) A&A, 2026, [arXiv:2601.07831](#)
9. Cosmoglobe DR2. VII. Towards a concordance model of large-scale thermal dust emission for microwave and infrared frequencies.
E. Gjerløw, R. M. Sullivan, R. Aurvik, et al. (Submitted) A&A, 2026, [arXiv:2601.07822](#)
8. Cosmoglobe DR2. V. Spatial correlations between thermal dust and ionized carbon emission in Planck HFI and COBE-DIRBE.
E. Gjerløw, R. M. Sullivan, R. Aurvik, et al. (Submitted) A&A, 2026, [arXiv:2601.07818](#)
7. Cosmoglobe DR2. III. Improved modelling of zodiacal light with COBE-DIRBE through global Bayesian analysis
M. San, A. Bonato, M. Galloway, et al. (Submitted) A&A, 2026, [arXiv:2408.11004](#)
6. RHINO: A large horn antenna for detecting the 21cm global signal.
P. Bull, A. El-Makadema, H. Garsden, et al. RASTI, 2025, [arXiv:2410.00076](#)
5. Bayesian estimation of diffuse radio foregrounds with large interferometric arrays.
K. A. Glasscock, University of Manchester, 2025, PhD Thesis
4. High-dimensional inference of radio interferometer beam patterns I: Parametric model of the HERA beams.
M. J. Wilensky, J. Burba, P. Bull, H. Garsden, **K. A. Glasscock**, et al. RASTI, 2024, [arXiv:2403.13769](#)
3. Disentangling the anisotropic radio sky: Fisher forecasts for 21cm arrays
Z. Zhang, P. Bull, **K. A. Glasscock**. MNRAS, 2024, [arXiv:2403.13768](#)
2. Sensitivity of Bayesian 21 cm power spectrum estimation to foreground model errors.
J. Burba, P. Bull, M. J. Wilensky, F. Kennedy, H. Garsden, **K. A. Glasscock**. MNRAS, 2024, [arXiv:2403.13767](#)
1. Statistical estimation of full-sky radio maps from 21cm array visibility data using Gaussian Constrained Realisations
K. A. Glasscock, P. Bull, J. Burba, H. Garsden, M. J. Wilensky. RASTI, 2024, [arXiv:2403.13766](#)

Conferences and workshops

- 31/08 - 18/09 2026 **Main organiser of Global End-to-End Analysis at All Frequencies workshop and conference.**
- 20/07 – 24/07 2026 **Simons Observatory Face-to-Face 2026**, Toronto, Canada. Collaboration meeting.
- 20/04 – 24/04 2026 **MeerKLASS busyweek 2026**, Stellenbosch, South Africa. Online participant in the collaboration meeting.
- 11/03/2026 **General seminar (invited) at ISAS/JAXA**, Sagamihara, Japan. Gave a 1 hr seminar to collaborators and the general institute on the AKARI re-analysis work I have been co-leading within the CMB&CO group at UiO.
- 10/09 2025 **21cm cosmology overview talk at Bayesian Cosmological Data Analysis course 2025**, University of Oslo, Norway. Presented a 45 min overview talk on the field of 21 cm cosmology, its advances and challenges to the course participants. I was also a general instructor in the two week course.
- 15/09 – 19/09 2025 **MeerKLASS busyweek 2025b**, Turin, Italy. Online participant in the collaboration meeting.
- 07/07 – 11/07 2025 **National Astronomy Meeting (NAM 2025)**, Durham, UK. Session organiser of “EDI Community Initiatives”. Presenting poster on PhD project. Contribution to the ScienceArt session. Awarded competitive fee waiver for the conference.
- 17/03 – 21/03 2025 **MeerKLASS busyweek 2025a**, Stellenbosch, South Africa. Online participant in the collaboration meeting.
- 21/10 – 25/10 2024 **Cosmo’24**, Kyoto University, Kyoto, Japan. Presented a poster.
- 16/09 – 18/09 2024 **Cosmoglobe 2024 meeting**, University of Oslo, Norway. Participated in the collaboration meeting.
- 02/09 – 13/09 2024 **Bayesian cosmological data analysis course 2024**, University of Oslo, Norway. Participated in the course, learned to use the end-to-end Commander pipeline, and presented the project done during the workshop.
- 19/05 – 24/05 2024 **URSI RASC 2024**, Gran Canaria, Spain. Presented 20 min talk.
- 06/02 – 07/02 2024 **"Science with the 21-cm line" Kavli Focus meeting**, Kavli institute, Cambridge, UK. Presented 20 min talk.
- 29/01 – 03/02 2024 **Aarhus University Cosmology Group**, Denmark. Invited to present a 45 min talk.
- 19/8 – 26/08 2023 **URSI GASS 2023**, Sapporo Conference Center, Japan. Presented 20 min talk.
- 03/07 – 07/07 2023 **National Astronomy Meeting (NAM 2023)**, Cardiff University, UK. Presented 15 min talk.
- 17/01 – 19/01 2023 **SKA Cosmology SWG annual meeting**, University of Manchester, UK. Participated in the annual meeting.
- 16/12 – 22/12 2022 **Aarhus University Cosmology Group**, Denmark. Invited to present a 30 min talk.
- 17/10 – 21/10 2022 **Global 21-cm meeting**, UC Berkeley, US. Presented a poster and a 10 min talk.
- 10/10 – 14/10 2022 **HERA annual meeting (HAM 2022)**, UC Berkeley, US. Presented a short 5 min poster talk for the collaboration.
- 13/09 – 16/09 2022 **ICIC Bayesian inference workshop**, Imperial College London, UK.

Teaching

- 09/2023 – 01/2025 **Graduate Teaching Assistant**, University of Manchester, UK
Teaching *General Physics lab* and *Introduction to Programming*. Small group (~8 students) lab demonstration and large group (~50-100 students) shared teaching. Conducted student interviews, guided and advised students, marked exams and assignments, and completed multiple teaching courses.
- 01/2022 – 08/2023 **Graduate Teaching Assistant**, Queen Mary University of London, UK
Teaching of *Our Universe*. Completed training and gained experience with classroom teaching.
- 03/2018 – 09/2020 **Student educator** at the Steno Museum and the Ole Rømer Observatory, Aarhus
Gave talks and tours of the observatory and shows at the planetarium. Completed training on teaching to a broad audience and designing tours. Took part in the design-process of the electromagnetism exhibition and in designing the guided tour/class related to it.

05/2017 – 03/2019 **Student educator** at The Mobile University, Aarhus University.
Planned, taught and managed classes (90 minutes) and workshops (5 hours) on cosmology at upper secondary schools. Completed training on outreach-teaching and on planning, structuring and conducting workshops.

EDI engagement and other extracurricular activities

- 06/2025 **Nominated for the President's Distinguished Achievement Award, Postgraduate Researcher of the year**, University of Manchester
Nominated amongst 13 other PGR students across the entire university.
- 01/2024 – 08/2025 **Dept. of Physics and Astronomy EDI Committee member**, University of Manchester
- 07 – 11/07/2025 **Organiser; "EDI Community Initiatives"**, National Astronomy Meeting, Durham, UK.
Proposed, organised and ran the sessions that focused on sharing best practises, spreading awareness and building connections between researchers, students and the general EDI community.
- 01/2024 – 05/2025 **Student lead of EquiTea** at JBCA, University of Manchester
Organised the department's EDI forum, now regular monthly sessions with engaged discussions that continue into the coffee break, often with invited speakers. Ensured signposting for the university's anti-harassment *Report & Support* channel. Secured funding (2025) to further increase engagement.
- 01/2023 – 08/2025 **Committee member of EquiTea** at JBCA, University of Manchester
- 08/06/2024 **Universally Manchester Community Fair**, University of Manchester, UK
Represented the department at the festival, engaging over 150 visitors in astronomy-related activities.
- 22/05/2023 **Pint of Science (invited)**, Manchester, UK. Gave 1 hr public engagement talk to the local community on 21-cm cosmology. Sold out event.
- 11/2021 – 07/2022 **Organising and hosting** the departments bimonthly Large Scale Structure meeting
- 01/2019 – 12/2019 **Director and head** of the biannual student revue (satirical sketches/theatre).
In charge of organising the writing process, rehearsals and performances as well as setting the manuscript, role list, directing, overseeing set- and costume design.
- 02/2018 – 12/2018 **Guide/tutor for undergrad students**, Mat/Fys-Tutorgruppen, Aarhus University.
Providing academic and social guidance for new students and organising induction days at the faculty of Science & Technology (*now; Faculty of Natural Science*).
- 02/2018 – 08/2018 **Board member** of the graduation committee (Den Ultrahemmelige Højtid) at Department of Physics and Astronomy, Aarhus University.
- 09/2016 – 08/2017 **Head of Public Relations and board member** of TÅGEKAMMERET.
Primary responsibility for communication, advertisement, creativity and outreach for the student organisation covering the Faculty of Science and Technology at Aarhus University.
- 09/2015 – 09/2016 **Event committee member (FU)** for TÅGEKAMMERET, Aarhus University.
Assisting the board of the organisation at events such as talks, pub quiz, students cafes and parties.
- 09/2014 – 12/2014 **Volunteer in Kenya** for the Danish organisation Save a Heart.
Assisted with establishing the now locally-run children's home for HIV-infected children: Loving Hands Safe House, Gilgil.
Teacher at Bondeni School in the slum of Gilgil and at AIC School, Gilgil.

Technical skills and interests

- Computing skills* High-performance computing (DiRAC), Linear solvers, Bayesian methods, Constrained realisations, Gibbs sampling
- Prog. languages* Python, $\LaTeX$, MatLab, C
- Libraries* NumPy, SciPy, Matplotlib, HEALPy, Astropy
- Dev. tools* Git, GitHub
- Research software* Hydra, CLASS, MontePython, CONCEPT
- Languages* Danish, English
- Hobbies* Bouldering, baking, art, singing, theatre